

Cisco Packet Tracer Project #02

VLAN & Trunking

Type: Networking Simulation **Difficulty:** Beginner–Intermediate **Tool:** Cisco Packet Tracer

1. Project Overview

This project simulates a small office network divided into IT and Finance. VLAN 10 is used for IT and VLAN 20 for Finance. Two Cisco 2960 switches are connected through a trunk so both VLANs can traverse one physical link. No router is used, so inter-VLAN communication is intentionally unavailable.

2. Objectives

Create VLANs; assign access ports; configure a trunk; verify same-VLAN communication across switches; verify VLAN isolation; and practice Cisco switch troubleshooting.

3. Addressing

Device	VLAN	IP Address
PC-01	10	192.168.10.10
PC-02	10	192.168.10.11
PC-03	10	192.168.10.12
PC-04	20	192.168.20.10
PC-05	20	192.168.20.11
PC-06	20	192.168.20.12
PC-07	10	192.168.10.13
PC-08	10	192.168.10.14
PC-09	20	192.168.20.13
PC-10	20	192.168.20.14

4. Port Configuration

SW1 Fa0/1–Fa0/3 → VLAN 10; Fa0/4–Fa0/6 → VLAN 20; Fa0/24 → trunk. SW2 Fa0/1–Fa0/2 → VLAN 10; Fa0/3–Fa0/4 → VLAN 20; Fa0/24 → trunk.

5. Verification & Testing

Test	Expected Result
PC-01 → PC-02	Success
PC-01 → PC-07	Success — VLAN 10 across trunk
PC-04 → PC-09	Success — VLAN 20 across trunk
PC-01 → PC-09	Request timed out — VLAN isolation

6. Technical Explanation

A VLAN creates a separate logical broadcast domain. Access ports carry traffic for a specific VLAN. A trunk carries multiple VLANs between switches using VLAN tagging such as IEEE 802.1Q. The trunk therefore allows VLAN 10 traffic to cross from SW1 to SW2 while preserving VLAN separation. Routing is not performed, so VLAN 10 cannot communicate directly with VLAN 20.

7. Troubleshooting Checklist

Use ``show vlan brief``, ``show interfaces trunk``, and ``show interfaces status``. Also check IP addresses, subnet masks, physical links, trunk interfaces, and VLAN existence on both switches.

8. Skills Demonstrated

VLAN segmentation, access ports, trunking, Layer 2 switching, IPv4 addressing, ICMP testing, Cisco IOS configuration, verification, and basic troubleshooting.

9. Next Project

Project #03 — Inter-VLAN Routing / Router-on-a-Stick: introduce Layer 3 routing so VLAN 10 and VLAN 20 can communicate through a router.